# BBS Pro

Importing Real Data into the Bar Bending Schedule

The @BBS drawing annotation and the CSV file — what they are, why they exist, and how to use them.

*Eurocode 2 / EN 1992-1-1 · EN ISO 3766 shape codes · Generated for BBS Pro*

## 1. Why this is needed — the problem

A CAD drawing (DXF or DWG) does not actually "know" anything about reinforcement. Internally it only stores geometry: lines, arcs, circles and pieces of text, each sitting on a named layer. When a structural engineer draws rebar, the engineer knows that a particular red polyline means "3 no. 16 mm bottom bars, 5800 mm long, mark B1" — but the file never records that meaning. It only stores a polyline and, perhaps, a text label that a human reads and interprets.

Because of this, no software can look at arbitrary CAD geometry and reliably work out the bar mark, diameter, shape and quantities. Two engineers will draw the same beam in two completely different ways. This is exactly why the original demo version of BBS Pro ignored the uploaded file and simply showed fixed, hard-coded data.

To make the tool work with real data, the bar information has to be written down in a structured form that the software can read. There are two honest ways to do that — and that is precisely what the @BBS annotation and the CSV file are. Both feed the same calculation engine (cut lengths, bending deductions, weights) and produce the same schedule.

## 2. The two ways to get real data in

| Method | Best for | What the user does |
|---|---|---|
| @BBS in a DXF | Designers working inside AutoCAD who want the schedule data to live with the drawing. | Add one @BBS text label per bar, then save the drawing as DXF. |
| CSV file | Anyone whose bar data is (or can be) in a spreadsheet — no CAD needed. | Fill in a spreadsheet with one row per bar and save it as CSV. |

*DWG note: AutoCAD's .dwg is a proprietary binary format that cannot be read in a web browser. Convert it to DXF first — see section 5.*

## 3. The @BBS annotation convention (DXF)

The @BBS convention is a small, fixed agreement on how to label bars inside the drawing so the software can read them. The app already reads the drawing's real layers automatically; the @BBS labels add the bar data on top of that.

### What the designer does, step by step (in AutoCAD)

- 1. Use the TEXT or MTEXT command to place a text label.
- 2. Make sure that label is on the correct layer (for example BEAMS-MAIN-BARS). The layer is read automatically and is used for grouping and colour.
- 3. Type the bar's data into the label using the @BBS format shown below.

- 4. Repeat for each bar type, then save the drawing as DXF (Save As -> AutoCAD DXF).

*The position of the label on the sheet does not matter — the app only reads the label's text, not where it sits. Keep one @BBS label per bar type.*

### The format

```
@BBS mark=B1; member=Beam B101; section=Ground Floor; shape=11;
     dia=16; A=4200; B=350; C=0; D=0; n=3; each=2; rem=Top hook bars
```

The label must begin with @BBS, followed by key=value pairs separated by semicolons. Keys:

| Key | Meaning |
|---|---|
| `mark` | Bar mark (required — a label with no mark is ignored) |
| `member` | Member or element the bar belongs to, e.g. Beam B101 |
| `section` | Section or floor, used for grouping, e.g. Ground Floor |
| `shape` | Shape code (see section 6) |
| `dia` | Bar diameter in millimetres |
| `A` `B` `C` `D` | Shape dimensions in millimetres (use only those the shape needs) |
| `n` | Number of identical members |
| `each` | Number of bars in each member |
| `rem` | Remarks (free text) |

A ready-made example drawing is bundled with the app at public/sample.dxf — open it in a text editor to see the exact structure. Total bars = n x each; the app then computes cut length, unit weight and total weight for you.

## 4. The CSV file

The CSV is the alternative input for people who do not want to touch the CAD drawing at all, or whose bar data already lives in a spreadsheet. A bar bending schedule is, at heart, just a table: one row per bar type. Many engineers and steel fixers already keep exactly that in Excel.

### What the user does

- Open Excel (or export from estimating / scheduling software).
- Create one row per bar type, with the column headers listed below in the first row.
- Save As -> CSV (Comma delimited).
- Import the CSV — the app does all the engineering (cut lengths, deductions, weights, totals).

### Recognised columns

*Header names are matched case-insensitively, and spaces or underscores are ignored (so "Bar Mark", "bar_mark" and "MARK" all work).*

| Column | Req. | Meaning |
|---|---|---|
| `mark` | Yes | Bar mark |
| `member` | - | Member / element |

| Column | Req. | Meaning |
|---|---|---|
| `section` | - | Section or floor (used for grouping) |
| `layer` | - | Layer name (used for grouping and colour) |
| `shapeCode` | - | Shape code (see section 6) |
| `dia` | - | Bar diameter in mm |
| `A B C D` | - | Shape dimensions in mm |
| `noOfMembers` | - | Number of identical members |
| `noOfBarsEach` | - | Bars per member |
| `remarks` | - | Free text |

### Minimal example

```
mark,member,section,layer,shapeCode,dia,A,B,C,D,noOfMembers,noOfBarsEach,remarks
B1,Beam B101,Ground Floor,BEAMS-MAIN-BARS,00,16,5800,0,0,0,3,4,Bottom main bars
L1,Beam B101,Ground Floor,BEAMS-LINKS,25,8,250,500,0,0,3,28,Links @200
```

A ready-made example is bundled at public/sample.csv.

## 5. Converting a DWG to DXF

A .dwg file cannot be imported directly. Convert it to DXF first — both options are free:

- In AutoCAD:  File -> Save As -> AutoCAD DXF (*.dxf).
- Without AutoCAD: use the free ODA File Converter (Open Design Alliance). Set the input to your DWG folder, output format to DXF, and convert.

Then import the resulting .dxf. Remember it still needs @BBS labels to carry the bar data (section 3).

## 6. Supported shape codes

| Code | Shape | Dimensions |
|---|---|---|
| `00` | Straight | A |
| `11` | L-Bar (1 bend) | A, B |
| `21` | Z-Bar (2 bends) | A, B, C |
| `25` | Closed Stirrup | A, B |
| `26` | Open Stirrup (U-top) | A, B, C |
| `32` | U-Bar | A, B |
| `33` | Cranked Bar | A, B, C |
| `37` | S-Bar (2 cranks) | A, B, C, D |
| `51` | Hook both ends | A |
| `99` | Non-standard / custom | A |

Cut lengths are computed per EN 1992-1-1 with a bending deduction of max(4 x dia, 20) mm per bend. Unit weights use a steel density of 7850 kg/m3. Shape codes follow EN ISO 3766.

## 7. The honest limitation

Tagging every bar with @BBS, or typing rows into a CSV, is manual work. That is the genuine limit of what is possible purely inside a web browser: the data has to come from somewhere structured. If you want the tool to read bars from drawings without manual tagging, that is a real feature, but it needs one of the following:

- An agreed drawing standard your firm already follows (for example, bars always on specific layers with callouts in a fixed text format). A parser can then be written for that exact format.
- A heavier backend that performs geometric interpretation, or an integration with a CAD / Revit API where rebar objects already carry their properties as real data.

*Until then, @BBS and CSV are the two reliable, transparent ways to get real schedules out of BBS Pro — and because all parsing happens in the user's own browser, uploaded drawings and schedules never leave their machine.*